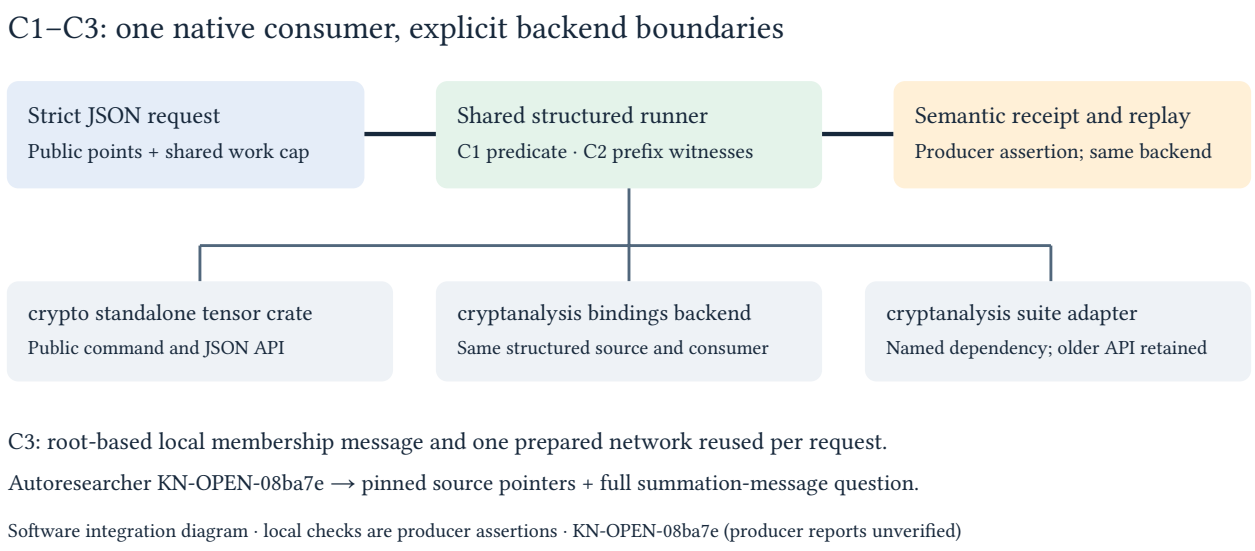
Structured prime-field IC consumer intake

2026-11-04 · KN-OPEN-08ba7e · producer reports, unverified

The intake links the implemented structured predicate, prefix-witness and local membership reuse components to one precise open question: a compact exact representation of the full constrained summation boundary, with construction, specialization and live-storage accounting. This is knowledge routing rather than a finding or technique promotion.



Source pins. crypto fff0182d77747b826fcc84443bde0a1c9a529509 and cryptanalysis 1c87f6887fe1047ce81a84bbf33cc97ed91e1cd7 carry the paired native consumer. The manifest integration.json pins source-file digests and commit URLs. The source PRs are crypto3419 and cryptanalysis1396; PR state is separate from the source pins. Software/check statements remain producer reports.

Consumer contract. A strict JSON request supplies public curve/subgroup and fibre parameters, query points and one explicit work budget. The consumer prepares the predicate/network once, returns counts and signed witnesses, and replays semantic answers with the same backend. Local membership messages are reused. The suite adapter names its separate backend instead of replacing the older tensor API. Receipts retain producer_assertion standing.

Open construction. Keep exact rational signed lifts, actual subgroup checks, all retained addition/pole charts and the ordered-tuple policy. Specify the boundary representation, coefficient algebra, precision, witness recovery and intermediate storage. Cost any all-target or all-tuple table explicitly and compare complete finite acceptance with the current direct reference before a separate performance protocol. These are construction requirements, not results.

Intake checks. The source and knowledge indexes are checked locally. The existing knowledge-entry metadata validator is applied to the new record only. The source-index unit panel is retained; an initial invocation named a missing knowledge-index unittest module and that failed output is retained too. Full ledger validation refuses the sparse checkout. No sparse override, experiment, evidence-strength change or research-state promotion is claimed.

Custody. TASK-20261104-5ee19d binds the exact archival paths, source commit, parent, record ID and file hashes. Archive-receipt.json records the existing GitRepositoryVerifier metadata check after the source commit. The process receipt is separate from scientific review.

Context. PKM2016, the pinned Hexagon prefix/weighted/locality reviews and the landed C1–C3 reference; existing shared-coefficient and algebraic-membership atlas leads remain open. The knowledge entry and README provide direct source links.